FINAL PRESENTATION

HIKINGGGGGGGG

QUICK RECAP

Background

Hiking participation expanded rapidly between 2014 and 2024, seeing an even larger surge during the COVID-19 pandemic.

This growth exposes more people to severe trail-induced injuries.

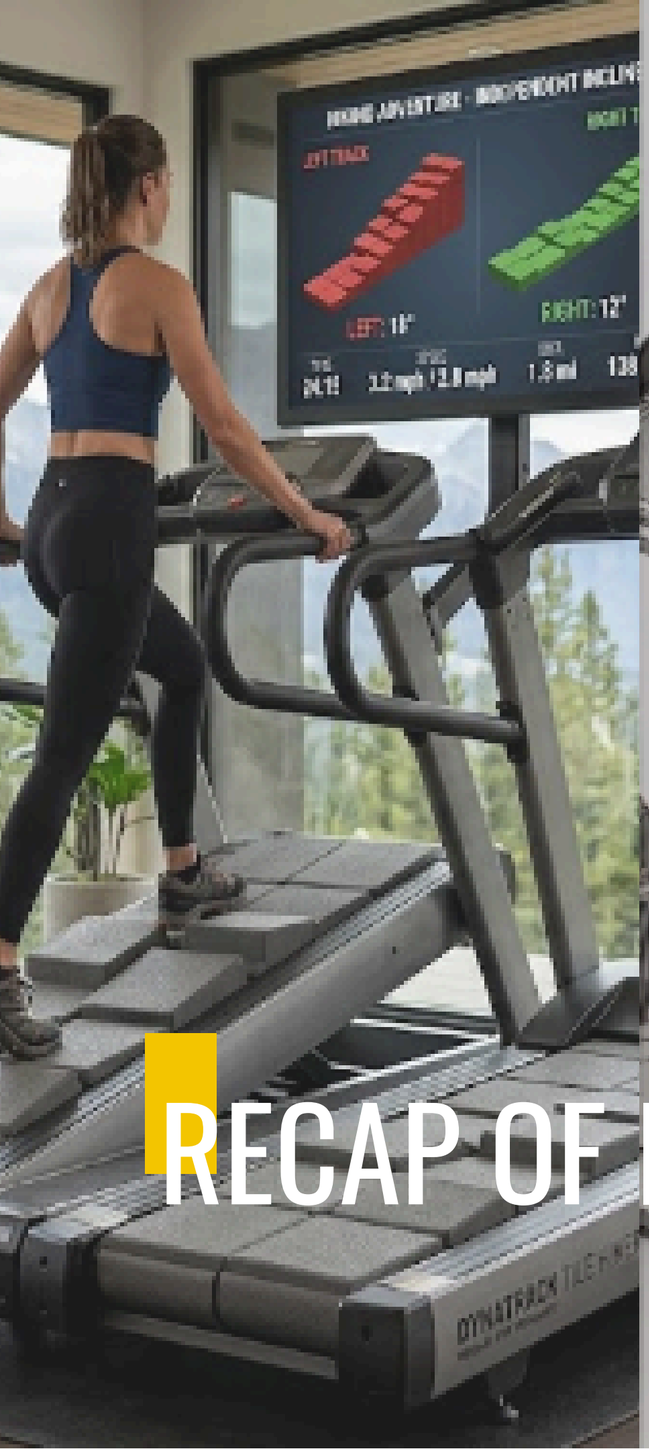
Injury Epidemic

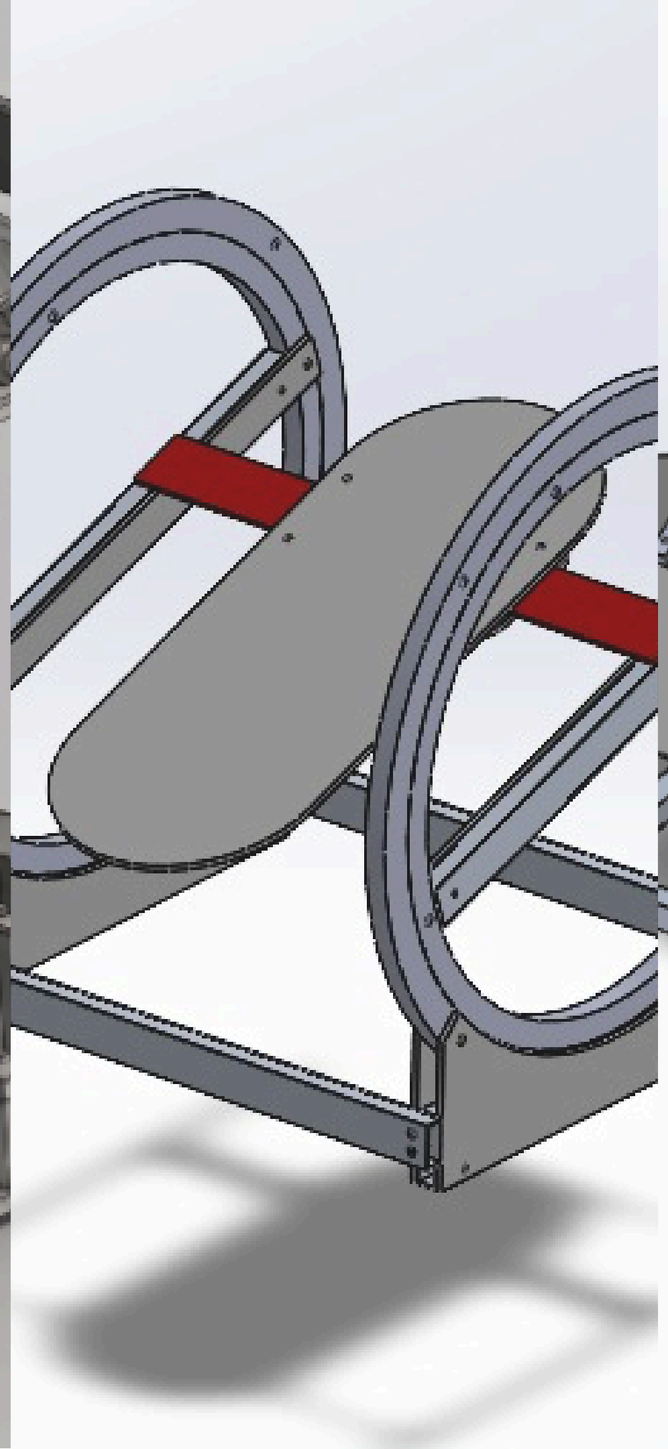
85%
ACUTE LATERAL SPRAINS

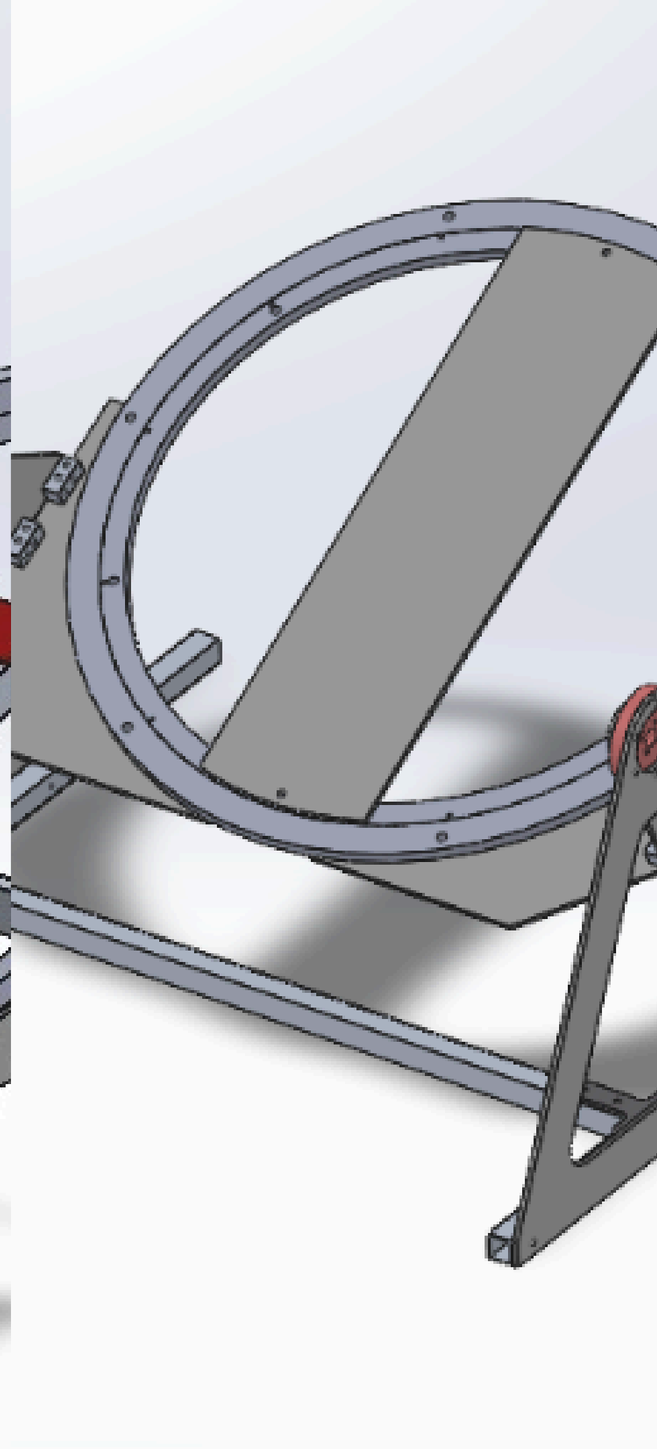
49%
DOWNHILL RACE INJURY RATE

THE PROBLEM STATEMENT

HIKERS NAVIGATING UNPREDICTABLE DOWNHILL TERRAIN FACE A PERFECT STORM FOR SUDDEN ANKLE INVERSION, SUFFERING SEVERE TIME-LOSS INJURIES THAT PREMATURELY END THEIR EXPEDITIONS AND CREATE LONG-TERM CHRONIC INSTABILITY.







RECAP OF IDEATION

TREADMILL

ideatation 01

R**eason:**

- Simulate the **real running or walking** motion.
- By adding **macro-tile architecture** we can **replicates the complex outdoor terrains.**

Why we banned:

- The mechanism is too **complicated.**
- we can’t make sure the **safety** of this product.

PEDAL TRAINER

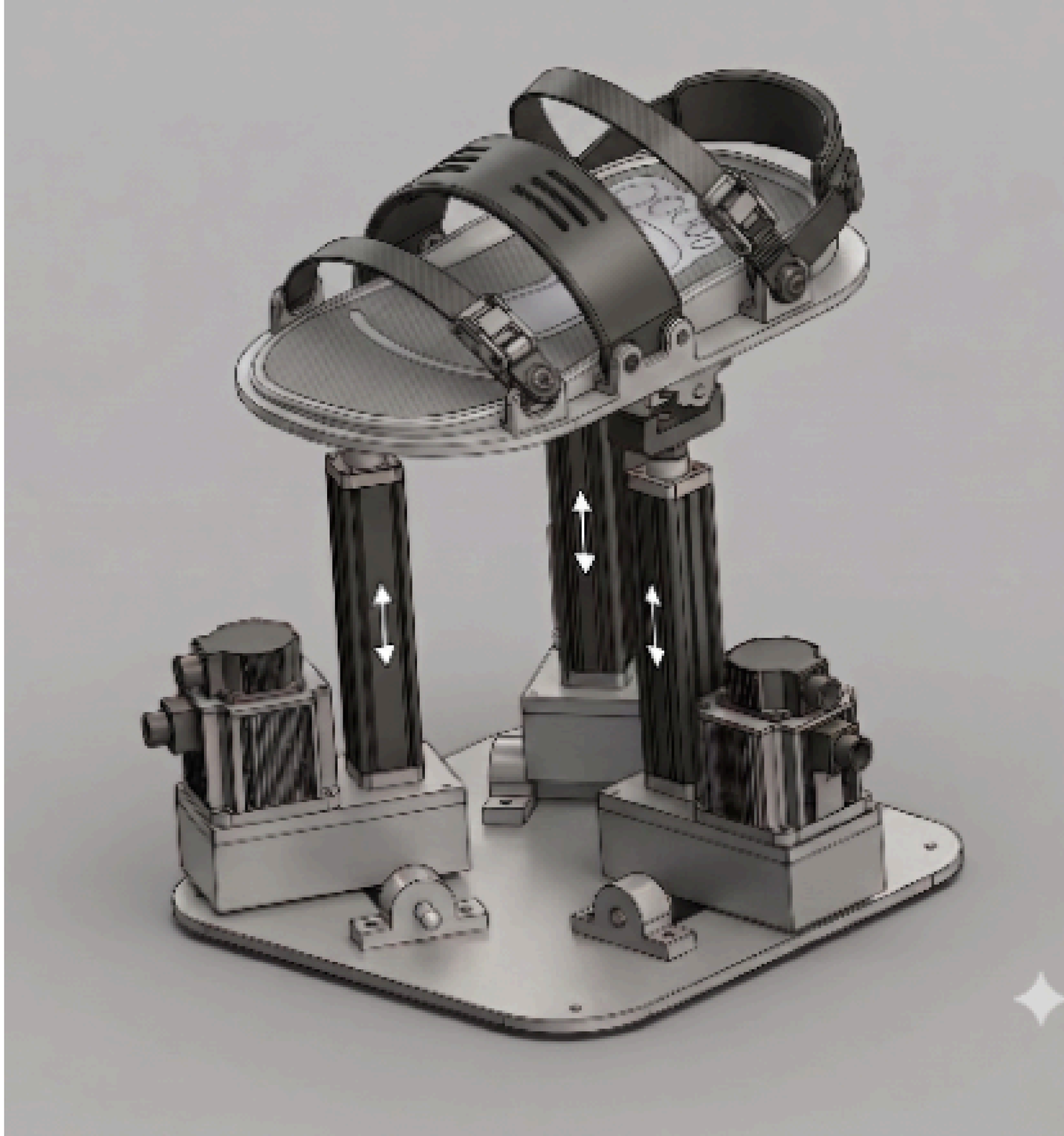
ideatation 02

Reason:

- Provide **Changable Load** for gym-style training.
- Shoes-like shape reduces the appearance of the blister.

Why we banned:

- The limited rotation range in Roll and Pitch axis.
- Unnecessary Yaw axis significantly increases mechanical and safety complexity.



TWISTING TRAINER

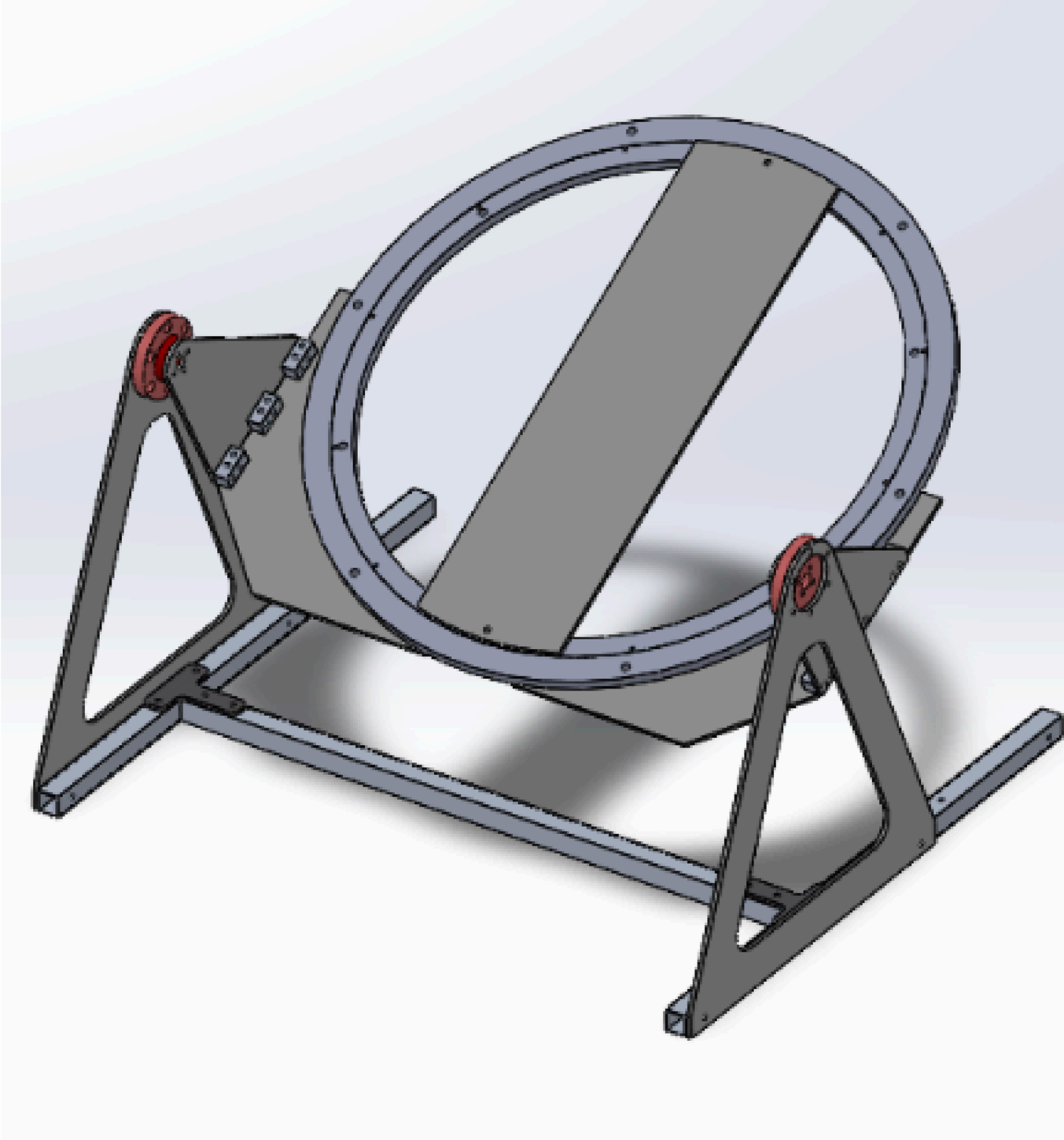
ideatation 03

Reason:

- Give out a way to train muscle through Yaw axis.
- Training muscle in Pitch axe with its gravity.
- Easy Mechanical structure.

Why we banned:

- The Circle may go out of the safe range to hurt the user.
- No roll axis training included.

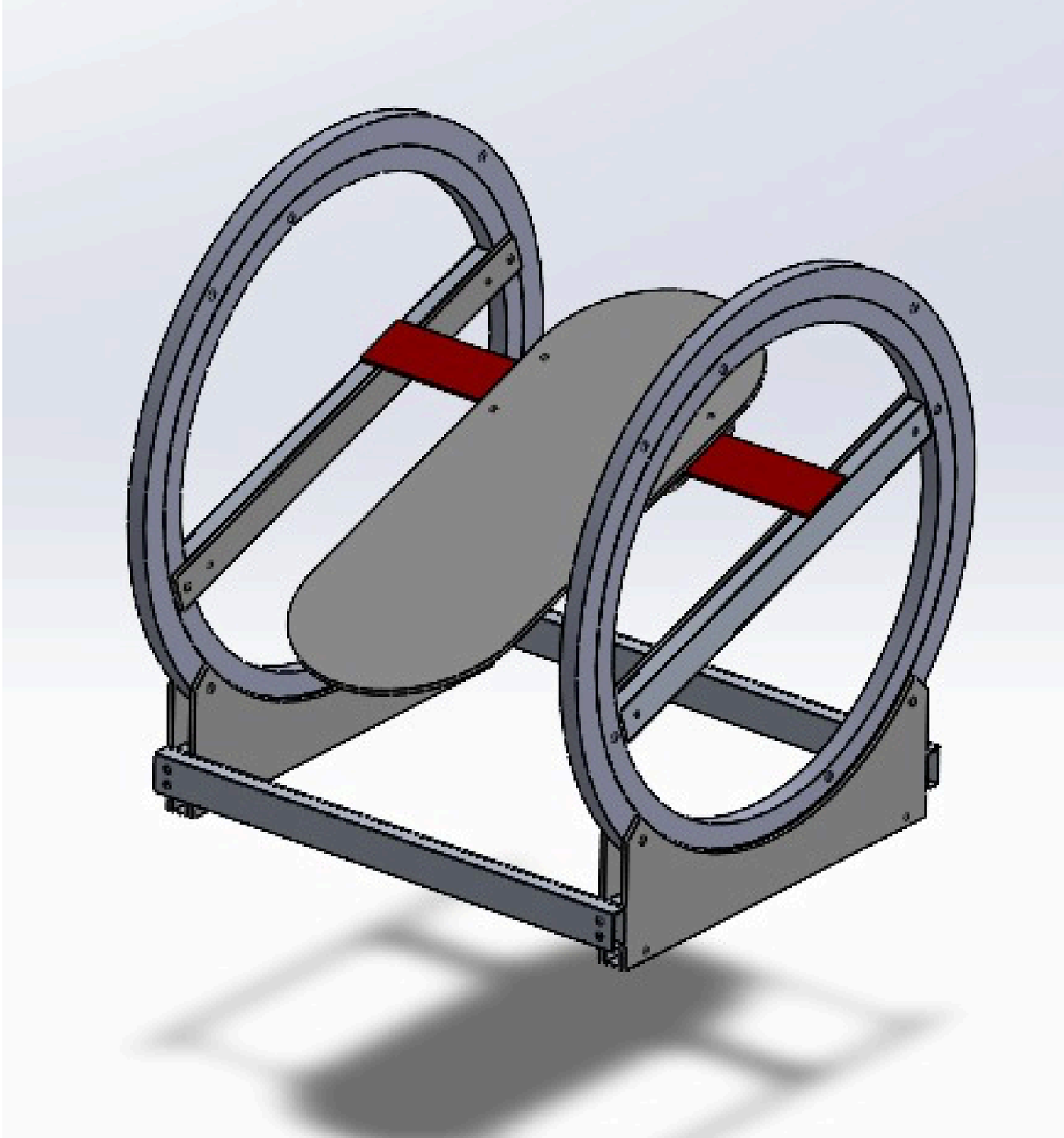


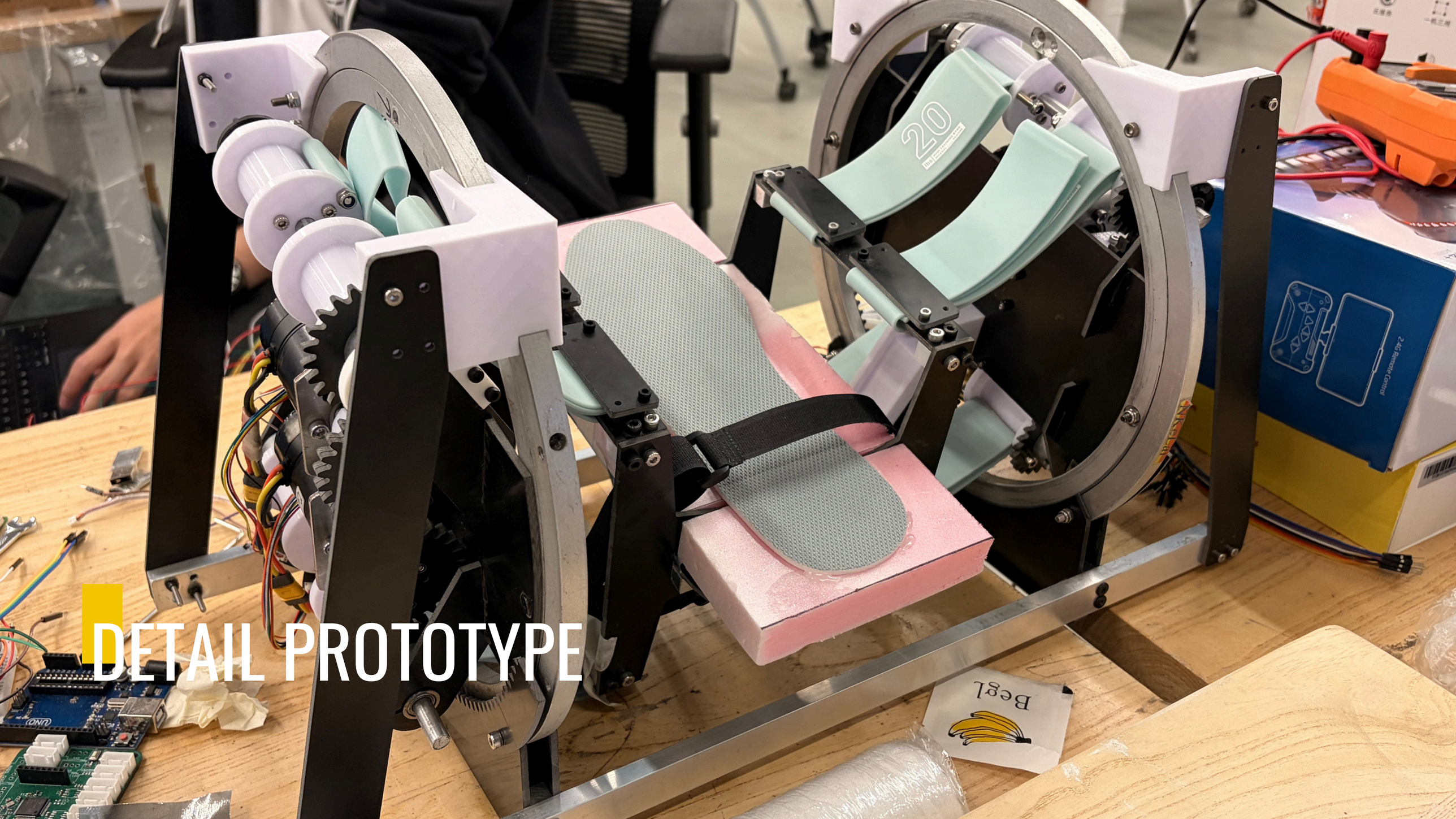
ACTUATION BELT

ideatation 04

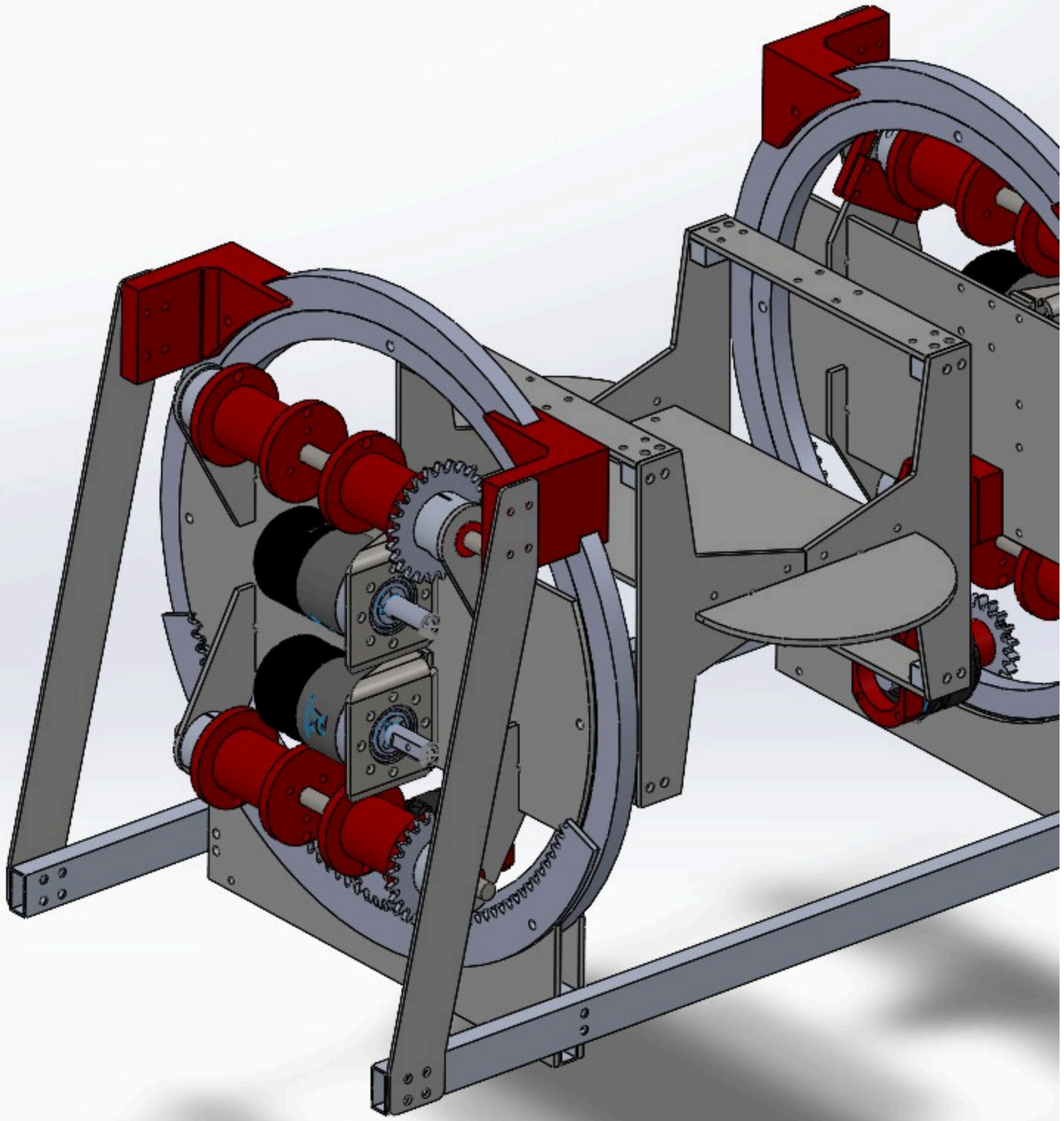
Reason:

- 3 DOFs with a large active space.
- Easy Mechanical to maintain and construction.
- Can apply changable external load.

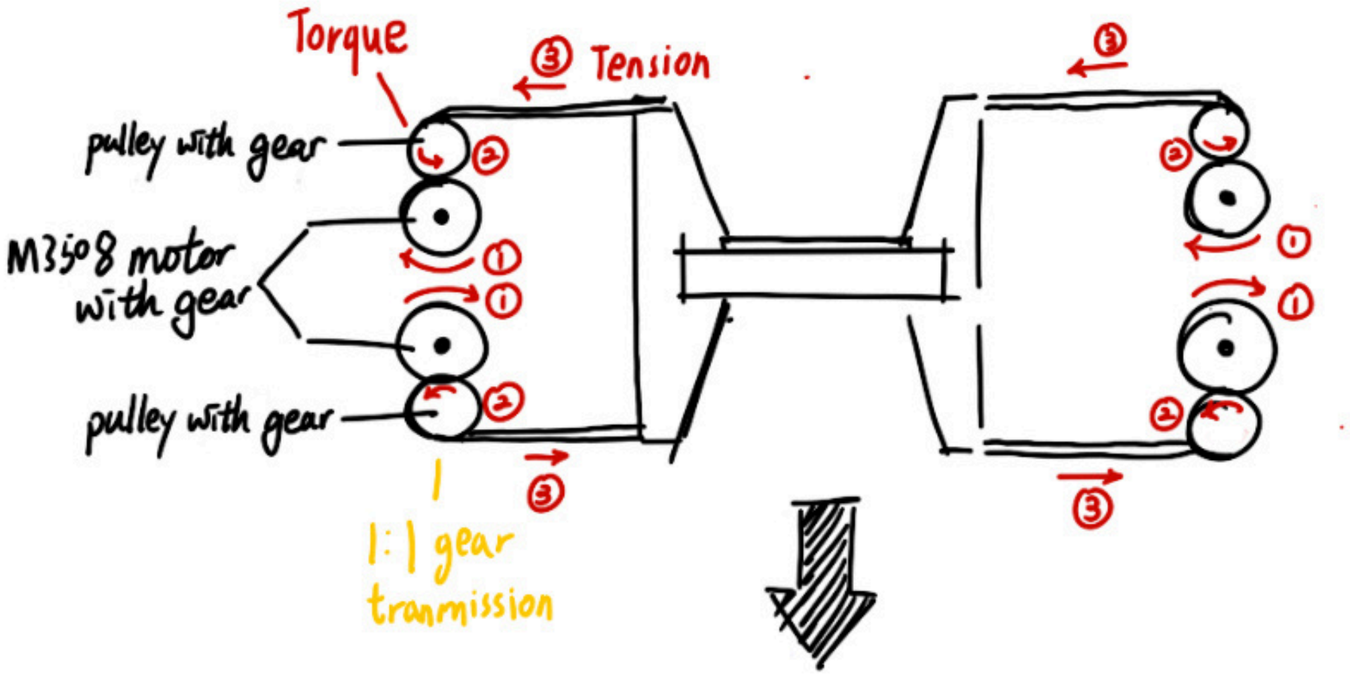


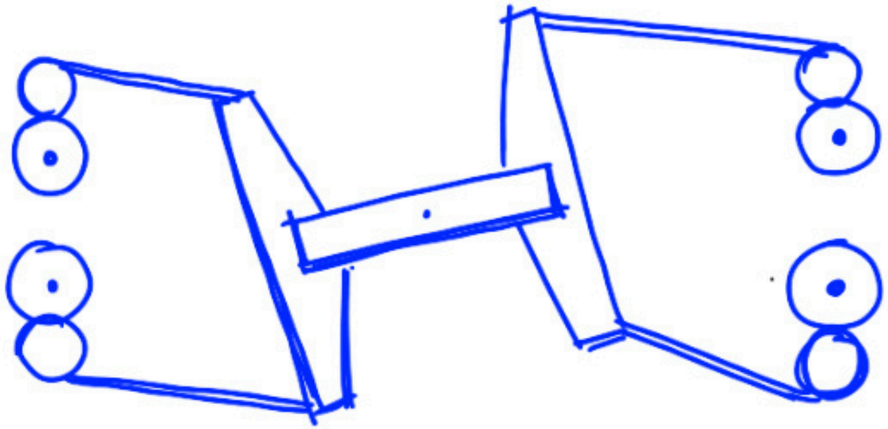


DETAIL PROTOTYPE



MECHANISM





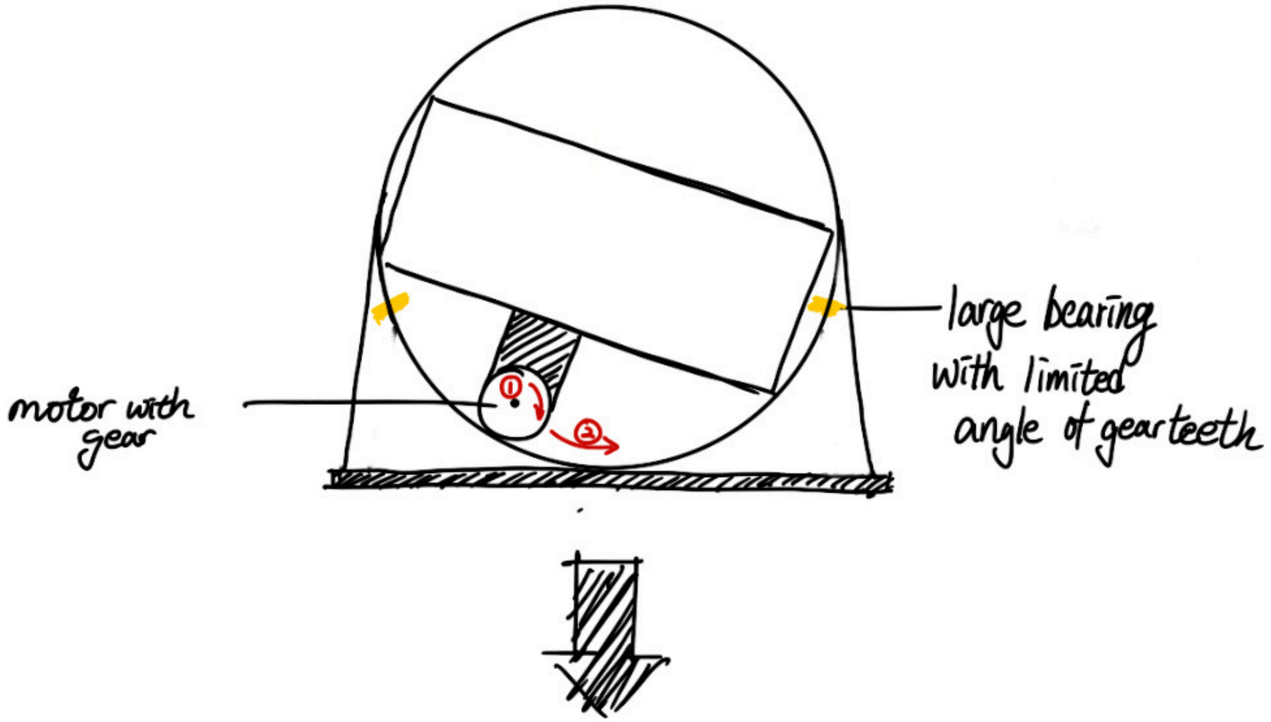
MECHANISM

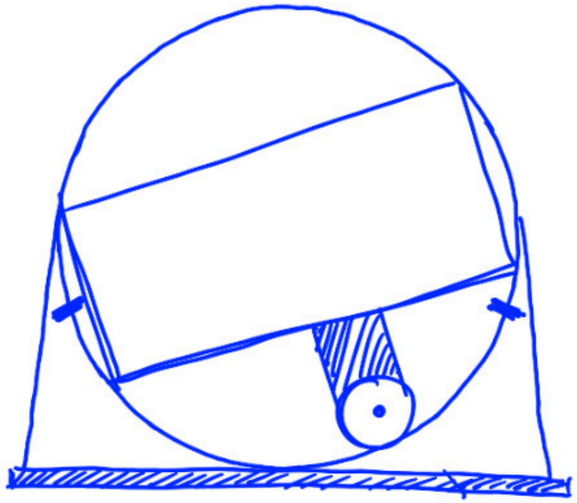
How it works - Roll

By controlling the motor, the gear will transfer torque to the pulley. Then this will produce a tension along the elastic belt to the connection part of the pedal.

By controlling the directions and torque of four different motor, we can produce a twist motion to the pedal just like how the picture show.

User can train their muscle by against the twisting motion of the pedal.





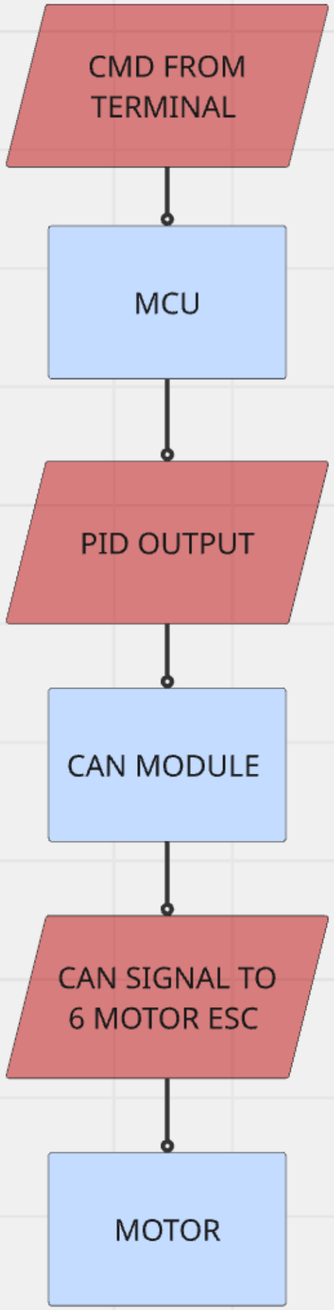
MECHANISM

How it works - Pitch

By controlling the motor, the gear will trun the whole middle module and then turn the pedal in the middle.

The whole system can turn smoothly because it is connect to a large bearing outside.

Adding a limited angle of gear teeth can make sure the motion of this system in an acceptable angle of human usage.

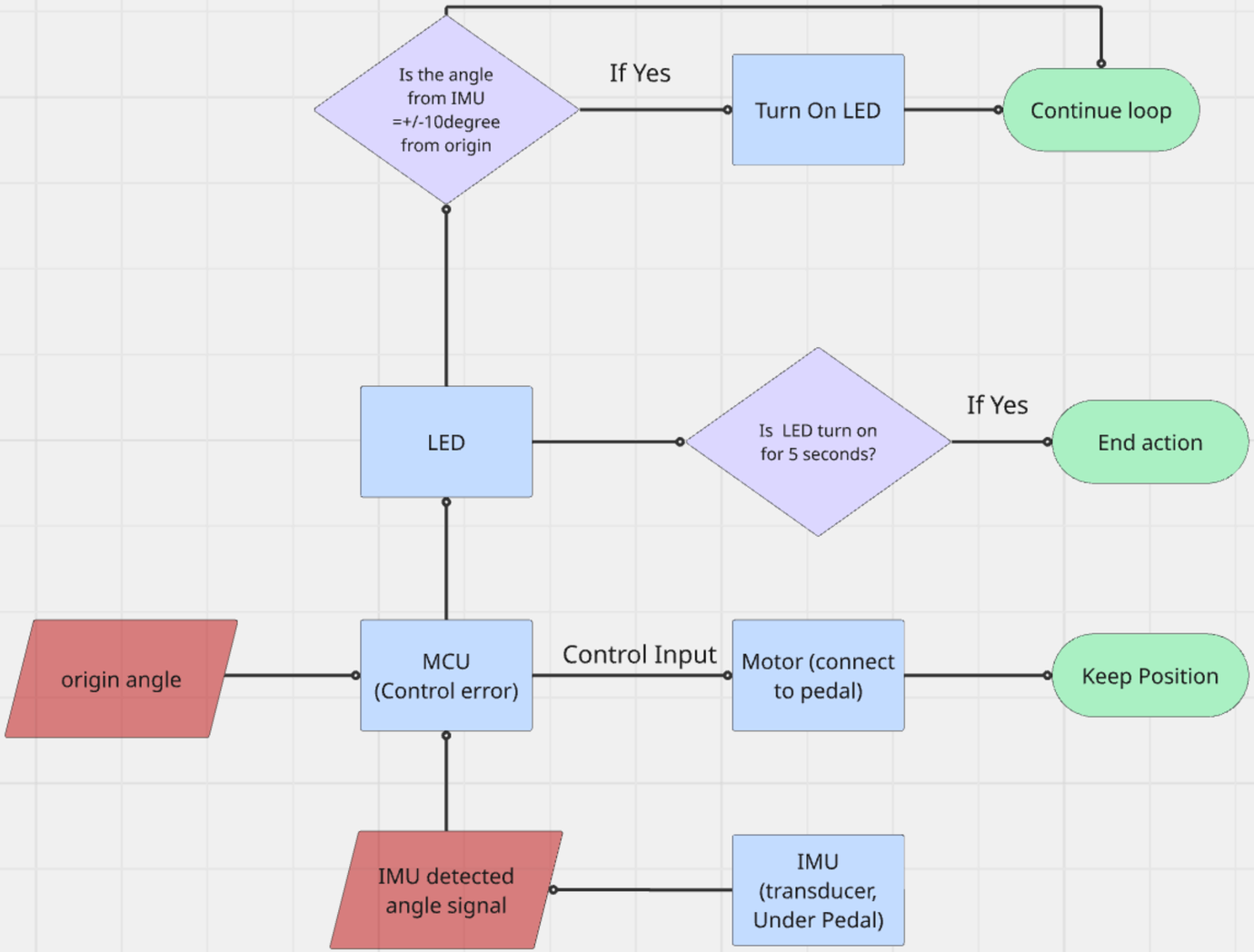


MOTOR SYSTEM

How it works

User input cmd from terminal , then can wireless control how the force of the motor, the direction of turning to let them train the muscle part they want.

With this motor system, we can have 3 degree of freedom in some case.

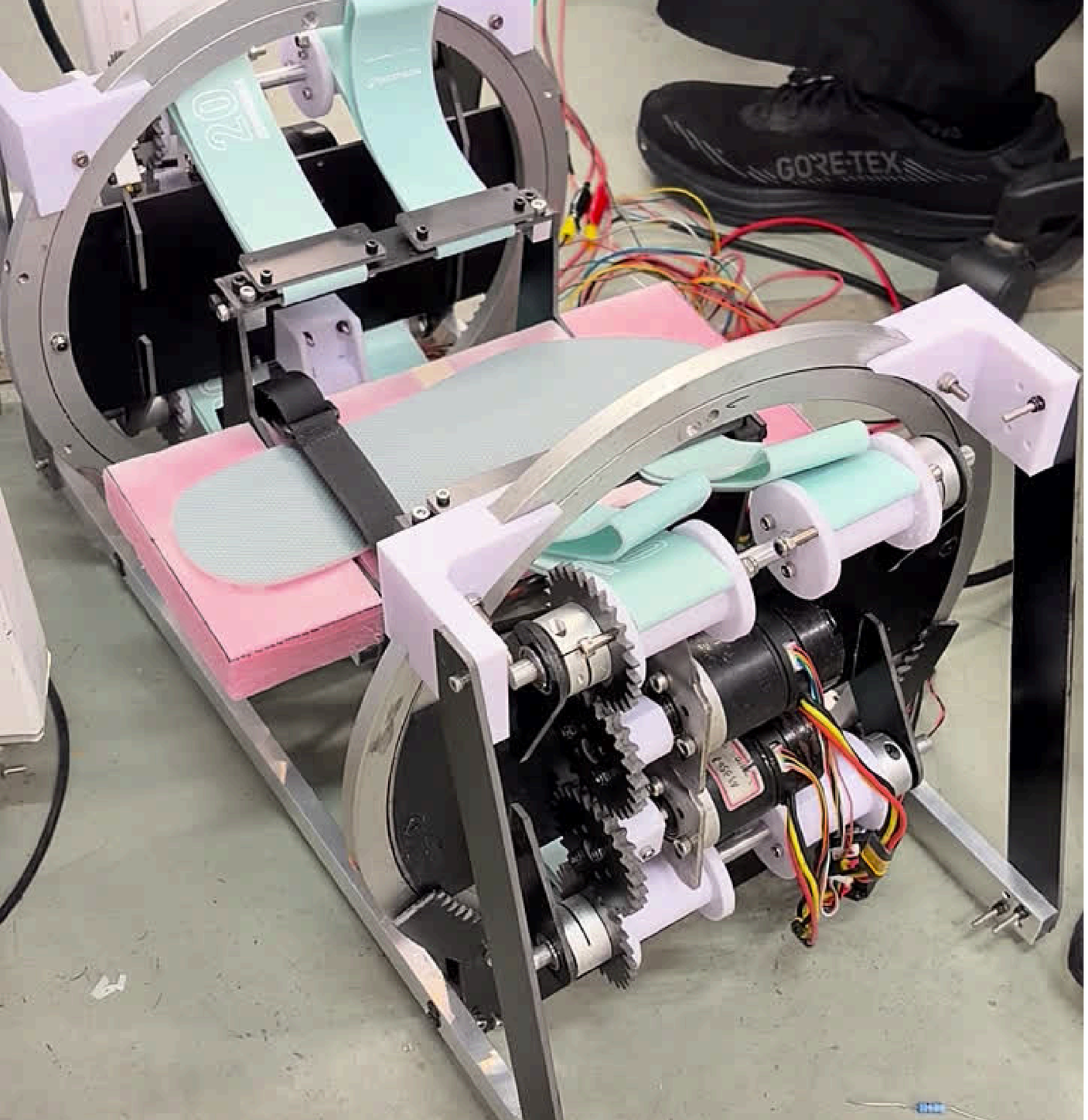


IMU

How it works

IMU is placed under pedal and use to detect the angle of pedal for all XYZ axis.

The system diagram shows how the control feedback loop of the IMU, MCU, motor and the pedal works together.
Also, we add two LED for user to know is the angle of their feet correct or not.



DEMO VID

TIMELINE

WEEK 1-3

- Develop an app for user control
- design hiking-like mode for the product
- Improve the mechanism and add more functions

WEEK 4-6

- BETTER PRODUCT OUTFIT
- DESIGN PEDAL FOR DIFFERENT FEET SIZE.
- REPLACE MATERIAL FOR ELECTRONS TO MINIMIZE THE SIZE.

WEEK 6-7

- FINISH THE BASIC FUNCTION AND OUTFIT
- DO USER TEST
- FIND IS THERE ANY OTHER POTENTIAL USERS

WEEK 7-8

- BASED ON THE FEEDBACK, DO EVALUATION
- DESIGN NEW FUNCTION OR FESTURE
- DETERMINE THE FUTURE PLAN

MB WE CAN TO DEVELOP THE PRODUCT LINE DEPTH TO FIT THE DEMAND OF DIFFERENT GROUP.

END

THANK YOU FOR YOUR LISTING

FUTURE PLAN

1. we will put an app for easy control (CV?, wt data we need to show?, what mode we want)
2. build interface on the product
3. Improve the mechanism and how to integrate all the motor to build different mode)
4. develop a better product outfit
5. change form for different usage scenario (sitting, standing?)
6. develop the pedal (can fit different user feet size)
7. Add more load function for different training scenario.

ABOUT US

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QUICK RECAP

MEET OUR TEAM

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+200

Our Team

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Jonathan Patterson

Margarita Perez